Question

The crystal structure of $\mathbf{MNO_3}$ is similar to a common crystal structure $\mathbf{X}$, and can be regarded as $\mathbf{M}$ and O jointly forming a cubic close-packed structure, with $\mathbf{N}$ filling the octahedral voids formed only by O.

However, unlike ordinary $\mathbf{X}$, in the close-packed layer formed by $\mathbf{M}$ and O, the position of $\mathbf{M}$ remains unchanged, but each equilateral triangle composed of O rotates clockwise or counterclockwise by the same and very small angle. It is known that the direction of rotation of every two $[\mathrm{O_3}]$ triangles sharing a vertex is different, and there are two such close-packed layers.

Select the correct option below.

A. If A, B, C represent one type of rotated close-packed layer, and A', B', C' represent another type of close-packed layer, then in the $[\mathbf{MNO_3}]$ structure, one stacking period of the close-packed layer is $\ldots A'BCA' \ldots$

B. The proper unit cell of this crystal contains 6 formula units, with one structural motif representing 1 formula unit.

C. Given that the average bond length of $\mathbf{N}-\mathrm{O}$ in the crystal is $191\ \mathrm{pm}$, the volume of the primitive unit cell is $3.86 \times 10^{-22}\ \mathrm{cm^3}$.

D. All $[\mathbf{NO_6}]$ octahedra are connected by sharing vertices, and each $[\mathbf{NO_6}]$ octahedron is connected to 3 $[\mathbf{NO_6}]$ octahedra.

E. In this crystal, the same atoms have the same chemical environment and the same spatial environment.

F. If $\mathbf{M}$ and O are regarded as close-packed layers, then all $\mathbf{N}$ fill into the octahedral voids composed only of O, with a filling fraction of $\frac{1}{3}$.

G. O exists and exists only in the middle of every two adjacent $\mathbf{N}$ atoms.

H. All other options are incorrect

Answer

Correct Answer: G

Detailed Explanation

According to the problem description, M and O together form a cubic closest packing, and N occupies the octahedral voids composed solely of O. This crystal structure is similar to the cubic perovskite structure, where X is the cubic perovskite, with M corresponding to Ti and N corresponding to Ca.

The stacking sequence with N at the vertices in the cubic system is: $\ldots ABCABCA\ldots$

In the close-packed layers formed by M and O, the positions of M remain unchanged, but each equilateral triangle composed of O is rotated by the same small angle, either clockwise or counterclockwise. It is known that every two adjacent O_3 triangles (within the same close-packed layer) sharing a vertex rotate in opposite directions. There are two such close-packed layers. To ensure that the two O_3 triangles forming the octahedral voids in adjacent layers rotate in the same direction, the two types of close-packed layers must alternate.

The stacking period of the close-packed layers in the MNO_3 structure is $\ldots AB'CA'BC'A\ldots$, which does not match the description of option A ($\ldots A'BCA'\ldots$), so option A is incorrect.

CHECKPOINT 1 PTS

The stacking period of the close-packed layers in the MNO_3 structure is $\ldots AB'CA'BC'A\ldots$, so option A is incorrect

Based on the stacking period, it can also be determined that a unit cell contains 6 N atoms, meaning it contains 6 formula units. The spatial environment is identical for every two adjacent layers, exhibiting translational symmetry. A structural motif consists of 2 formula units, which contradicts option B's description of a structural motif representing 1 formula unit. Thus, option B is incorrect.

CHECKPOINT 1 PTS

The proper unit cell of this crystal contains 6 formula units, and a structural motif represents 2 formula units, so option B is incorrect

It is known that the average $N-O$ bond length in this crystal is $191pm$.

Let the edge length of the NO_6 octahedron be d. Then, the height of the octahedron is $h=\frac{\sqrt{6}d}{3}$.

It is straightforward to show that $a=2d$ and $c=6h=2\sqrt{6}d$. Therefore,

$\frac{c}{a}=\sqrt{6}$

CHECKPOINT 1 PTS

$\frac{c}{a}=\sqrt{6}$

$$d=\sqrt{2}d(N-O)=270pm$$

CHECKPOINT 1 PTS

$$d=\sqrt{2}d(N-O)=270pm$$

The volume of the proper unit cell is

$$(270\times2)^2\times2\sqrt{6}\times270\times\frac{\sqrt{3}}{2}\times10^{-30}=3.34\times10^{-22}cm^3$$

This contradicts option C's description of the proper unit cell volume as $3.86\times10^{-22}cm^3$, so option C is incorrect.

CHECKPOINT 1 PTS

The volume of the proper unit cell is $3.34 \times 10^{-22} cm^3$, so option C is incorrect

All NO_6 octahedra are vertex-sharing, and each NO_6 octahedron is connected to 6 other NO_6 octahedra. This contradicts option D's description of each NO_6 octahedron being connected to 3 others, so option D is incorrect.

CHECKPOINT 1 PTS

All NO_6 octahedra are vertex-sharing, and each NO_6 octahedron is connected to 6 other NO_6 octahedra, so option D is incorrect

From the environment of oxygen atoms in adjacent layers, it can be inferred that atoms of the same type in this crystal have identical chemical environments but different spatial environments. This contradicts option E's description of identical chemical and spatial environments for the same atoms, so option E is incorrect.

CHECKPOINT 1 PTS

In this crystal, atoms of the same type have identical chemical environments but different spatial environments, so option E is incorrect

If M and O are treated as close-packed layers, all N atoms occupy the octahedral voids composed solely of O. The octahedral voids composed solely of O account for $\frac{1}{4}$ of all octahedral voids, resulting in an occupancy rate of $\frac{1}{4}$. This contradicts option F's description of an occupancy rate of $\frac{1}{3}$, so option F is incorrect.

CHECKPOINT 1 PTS

If M and O are treated as close-packed layers, all N atoms occupy the octahedral voids composed solely of O, with an occupancy rate of $\frac{1}{4}$, so option F is incorrect

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N occupies the center of the octahedron, and O forms the vertices of the octahedron. Since N only exists in the octahedral voids formed by O, it follows that O exists solely between every two adjacent N atoms, meaning option G is correct.

CHECKPOINT 1 PTS

O exists solely between every two adjacent N atoms, so option G is correct